Week 3 Update

Michael Fang

March 2026

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Main topics:

- Why spreadability didn't work for BT (it converges very slowly)
- Found 504 new chains at $\alpha = 2$ from cubic matrices
- Generalized metrics Λ_{II}, Λ_{III} for Classes II/III
- $|\det M|$=1 matrices cannot have $2 < \alpha < 3$, but $|\det M|$>1 can
- Silver $\bar{\Lambda} = 1/4$ conjecture, metallic-mean limit $\to 1/3$

Recap

Last week: computed $\bar{\Lambda}$ for 9 patterns (Fibonacci through stealthy), confirmed α from eigenvalue formula, identified Classes I/II/III behavior.

Since last meeting:

- Determined why spreadability failed for Bombieri-Taylor
- Tested convergence of $\bar{\Lambda}$ and spreadability with system size
- Explored 3×3 substitution matrices (39,304 with row sums ≤ 4)
- Explored 4×4 matrices (24M with row sums ≤ 3)
- Defined generalized metrics Λ_{II}, Λ_{III} for Classes II and III
- Proved $2<\alpha<3$ inaccessible when $|\det M| = 1$
- Found chain with $|\det M| = 2$ at $\alpha = 2.071$ (first in gap)
- Silver $\bar{\Lambda} = 1/4$ conjecture, metallic-mean limit $\to 1/3$

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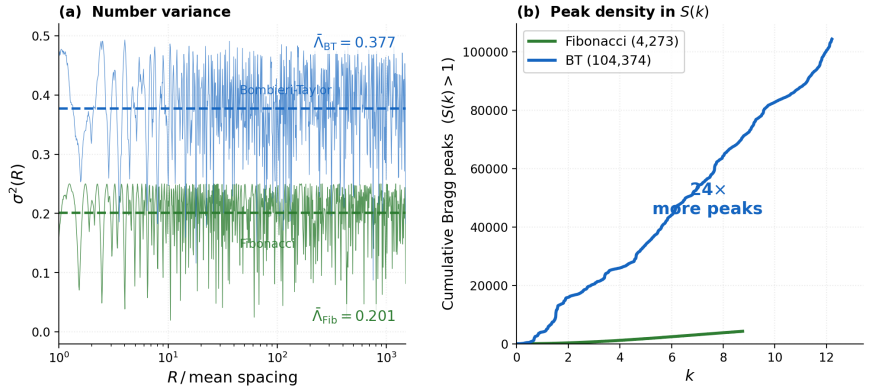
- Silver $\bar{\Lambda} = 1/4$ conjecture, metallic-mean limit $\to 1/3$

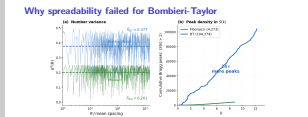
Week 2 results:

Pattern	α (meas)	α (exp)	$\bar{\Lambda}$
Fibonacci	3.05	3	0.200
Silver	2.99	3	0.250
Bronze	2.99	3	0.282
URL (a=1)	—	2	0.333
Stealthy χ=0.3	—	—	0.357
BT	1.545	1.545	0.377
Period-Doubling	1.0	1	—
0222 chain	0.64	0.639	—

Why spreadability failed for Bombieri-Taylor

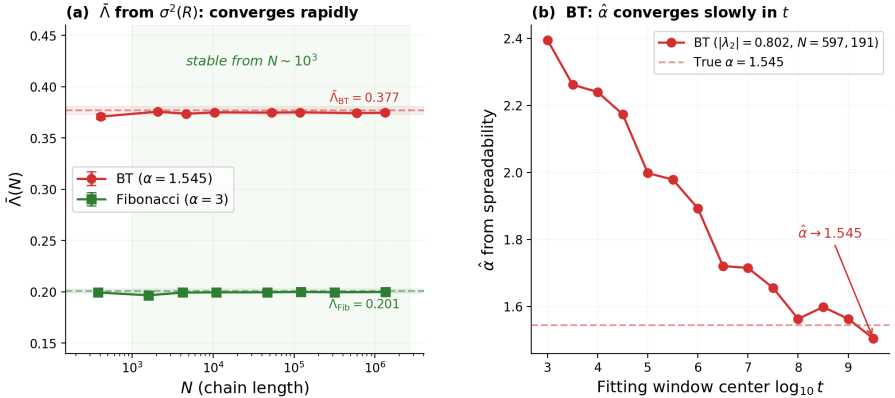


1. Large $|\lambda_2| = 0.802$ (vs. 0.618 for Fib) means correction terms in the spreadability decay slowly. The true power law only shows up at very large t ($> 10^8$). **2.** BT has 24× more Bragg peaks than Fibonacci (10^5 vs. 4,300), causing oscillations that bias any slope fit.

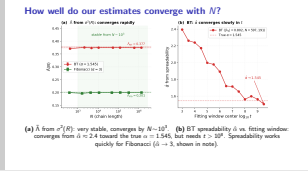


Spreadability converges to $\alpha = 1.545$ for BT, but slowly ($\hat{\alpha} = 2.26$ at $t \in [10^2, 10^5]$, 1.98 at $[10^4, 10^7]$, 1.51 at $[10^8, 10^{11}]$).
Original window ($t \in [10^2, 10^6]$) gave $\hat{\alpha} \approx 2.2$.
Large $|\lambda_2| = 0.802$ means correction terms decay very slowly, which messes with spreadability over much longer times.

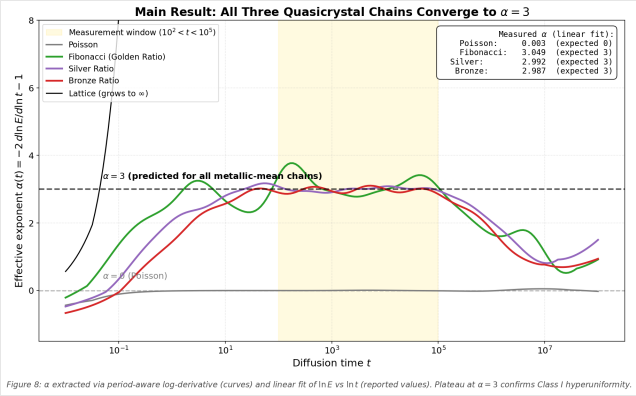
How well do our estimates converge with N?



(a) $\bar{\Lambda}$ from $\sigma^2(R)$: very stable, converges by $N \sim 10^3$. **(b)** BT spreadability $\hat{\alpha}$ vs. fitting window: converges from $\hat{\alpha} \approx 2.4$ toward the true $\alpha = 1.545$, but needs $t > 10^8$. Spreadability works quickly for Fibonacci ($\hat{\alpha} \to 3$, shown in note).



Fibonacci spreadability works fine ($\hat{\alpha} \to 3$ at moderate t):



Exploring cubic substitution matrices

Generated all 3×3 matrices with $|\det M| = 1$ and $|\lambda_i| < 1$ for $i \geq 2$, row sums ≤ 4:

$$39{,}304 \xrightarrow{|\lambda_i|<1} 4{,}882 \xrightarrow{\text{complex pair}} 504$$

All 504 give $\alpha = 2$ exactly.

For any 3×3 matrix with $|\det M| = 1$ where λ_2, λ_3 are a complex conjugate pair:

$$\underbrace{|\det M|}_{=1} = \lambda_1 |\lambda_2|^2 \;\Rightarrow\; |\lambda_2| = \frac{1}{\sqrt{\lambda_1}}$$

$$\Rightarrow\; \alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1} = 2 \quad \text{(exactly, for any } \lambda_1\text{)}$$

The remaining 4,378 matrices give $\alpha = 3$.

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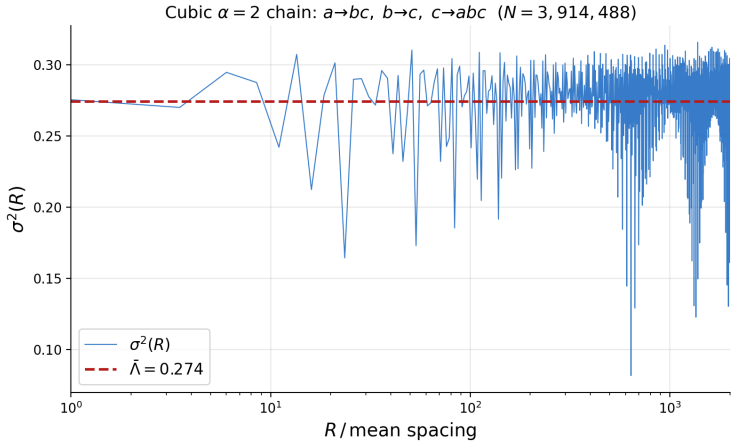
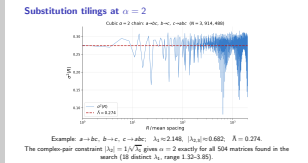
$|\lambda_i| < 1$ for all eigenvalues except λ_1 ensures pure Bragg diffraction (no diffuse background).
Complex pair: $\det M = \lambda_1 |\lambda_2|^2 = 1 \Rightarrow |\lambda_2| = 1/\sqrt{\lambda_1}$, $\alpha = 2$ for any λ_1.
When all eigenvalues are real, all matrices with row sums ≤ 4 give $\alpha = 3$

Substitution tilings at $\alpha = 2$



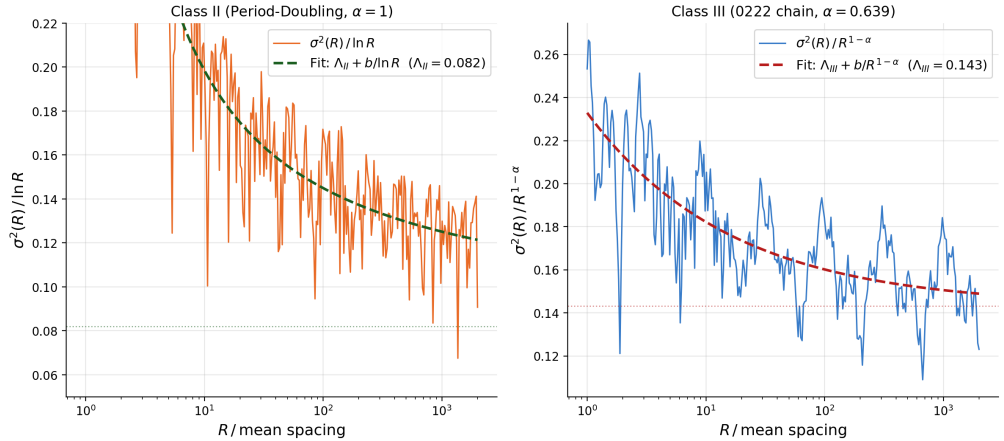
Example: $a \to bc,\ b \to c,\ c \to abc$; $\lambda_1 \approx 2.148$, $|\lambda_{2,3}| \approx 0.682$; $\bar{\Lambda} = 0.274$.
The complex-pair constraint $|\lambda_2| = 1/\sqrt{\lambda_1}$ gives $\alpha = 2$ exactly for all 504 matrices found in the search (18 distinct λ_1, range 1.32–3.85).



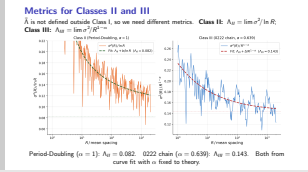
This example is just one of 504. $\bar{\Lambda}$ varies a lot across the 504 matrices (0.26 to 1.7), even between those sharing the same λ_1.

Metrics for Classes II and III

$\bar{\Lambda}$ is not defined outside Class I, so we need different metrics. **Class II:** $\Lambda_{II} = \lim \sigma^2/\ln R$; **Class III:** $\Lambda_{III} = \lim \sigma^2/R^{1-\alpha}$



Period-Doubling ($\alpha = 1$): $\Lambda_{II} = 0.082$. 0222 chain ($\alpha = 0.639$): $\Lambda_{III} = 0.143$. Both from curve fit with α fixed to theory.



Figures plot $\sigma^2/\ln R$ (Class II) and $\sigma^2/R^{1-\alpha}$ (Class III). If the scaling is correct, these ratios should flatten to a constant at large R.

Fit: $\sigma^2 = \Lambda_{II} \ln R + b$, so $\sigma^2/\ln R = \Lambda_{II} + b/\ln R$ (same idea for Class III, with α fixed to the theoretical value).

Updated catalog

Pattern	Class	α	$\bar{\Lambda}$
Integer Lattice	I	—	$1/6 = 0.167$
Fibonacci	I	3	0.201
Silver	I	3	$1/4 = 0.250$
Cubic α=2	I	2	0.274
Bronze	I	3	0.282
Copper	I	3	0.293
Nickel	I	3	0.310
URL ($a = 1$)	I	2	$1/3 = 0.333$
Stealthy χ=0.3	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Stealthy χ=0.2	I	—	0.526
Period-Doubling	II	1	($\Lambda_{II} = 0.082$)
0222 chain	III	0.639	($\Lambda_{III} = 0.143$)

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Matrices with $2 < \alpha < 3$

Can we find $\alpha \in (2, 3)$ at higher ranks? Searched 4×4 matrices too:

Rank	Matrices tested	$\lvert\lambda_i\rvert < 1$	$\alpha \in (2, 3)$
3×3 (row sums ≤ 4)	39,304	4,882	**0**
4×4 (row sums ≤ 3)	24M	1.36M	**0**

When $|\det M| = 1$, the eigenvalues are constrained:

- Rank 3 with complex pair: $\lambda_1 |\lambda_2|^2 = 1$ forces $\alpha = 2$ (previous slide)
- Rank ≥ 4 with complex pair: extra small eigenvalues force $|\lambda_2|$ larger, giving $\alpha < 2$
- All real eigenvalues: search finds only $\alpha = 3$

No substitution matrix with $|\det M| = 1$ can produce $2 < \alpha < 3$.

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Matrices with $2 < \alpha < 3$ (cont.)

Because $|\det M| = 1$ forces $\alpha \le 2$, we need $|\det M| > 1$ to get $\alpha \in (2,3)$. For any 2×2 matrix with real eigenvalues and $|\lambda_2| < 1$,

$$\boxed{\alpha = 3 - \frac{2\ln|\det M|}{\ln \lambda_1}}$$

$\lvert\det M\rvert$	Result	Example
1	$\alpha = 3$	metallic means
$(1, \sqrt{\lambda_1})$	$\alpha \in (2,3)$	**28 found ($\lvert\det\rvert=2$)**
$\sqrt{\lambda_1}$	$\alpha = 2$	boundary

28 examples found; all $|\det| = 2$, $\alpha \in (2.0,\ 2.2)$.

Smallest example ($|\det M| = 2$):

$$M = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}, \quad a \to b, \quad b \to aabbbb$$

$$\lambda_1 = 2 + \sqrt{6} \approx 4.449$$

$$|\det M| = 2 < \sqrt{\lambda_1} \approx 2.11$$

$$\alpha = 3 - \frac{2\ln 2}{\ln(2+\sqrt{6})} = \mathbf{2.071}$$

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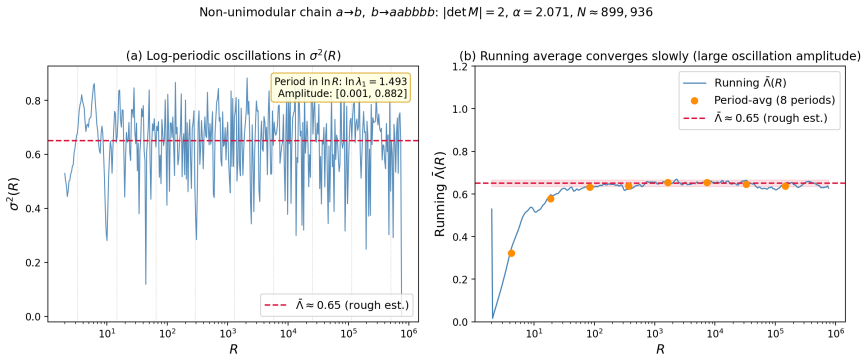
To get $\alpha \in (2,3)$: need $1 < |\det M| < \sqrt{\lambda_1}$.
Because det is an integer, only $|\det| = 2$ works (and only when $\lambda_1 > 4$).
This limits α to about 2.0–2.2. Getting closer to 3 would need a non-integer det.

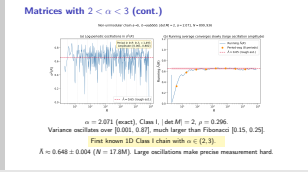
Matrices with $2 < \alpha < 3$ (cont.)



$\alpha = 2.071$ (exact), Class I, $|\det M| = 2$, $\rho = 0.296$.
Variance oscillates over [0.001, 0.87], much larger than Fibonacci [0.15, 0.25].

First known 1D Class I chain with $\alpha \in (2, 3)$.

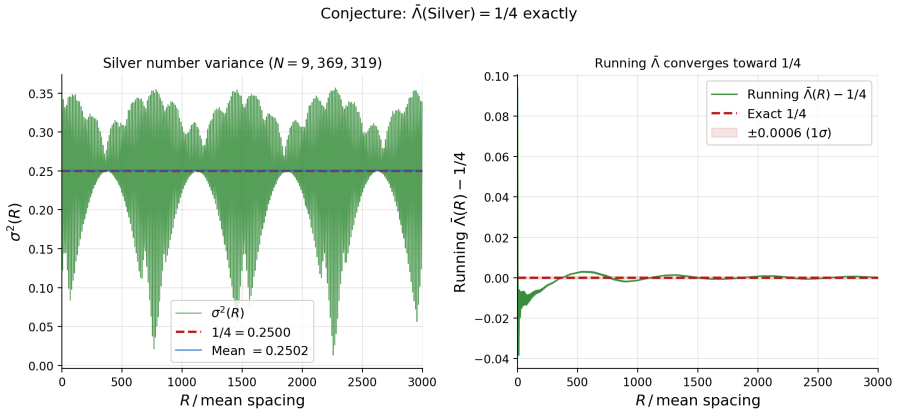
$\bar{\Lambda} \approx 0.648 \pm 0.004$ ($N = 17.8$M). Large oscillations make precise measurement hard.



Period averages still drifting up (0.643, 0.646, 0.652), so the true value might be a bit higher than 0.648.

The figures use $N = 900$k, the 0.648 ± 0.004 on the slide is from a larger $N = 17.8$M run.

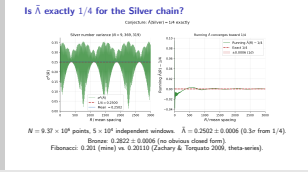
Is $\bar{\Lambda}$ exactly $1/4$ for the Silver chain?



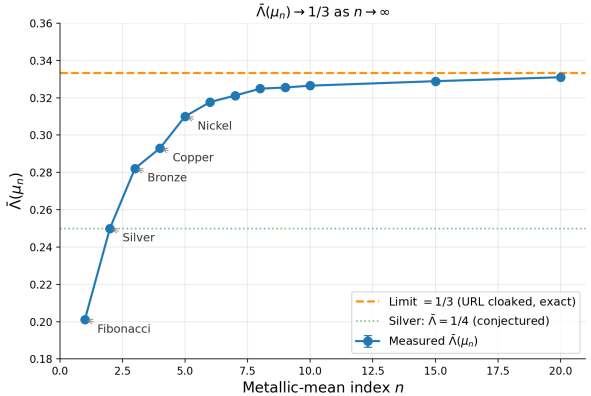
$N = 9.37 \times 10^6$ points, 5×10^4 independent windows. $\bar{\Lambda} = 0.2502 \pm 0.0006$ (0.3σ from $1/4$).

Bronze: 0.2822 ± 0.0006 (no obvious closed form).

Fibonacci: 0.201 (mine) vs. 0.20110 (Zachary & Torquato 2009, theta-series).



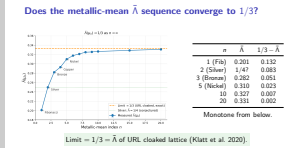
Does the metallic-mean $\bar{\Lambda}$ sequence converge to $1/3$?



n	$\bar{\Lambda}$	$1/3 - \bar{\Lambda}$
1 (Fib)	0.201	0.132
2 (Silver)	1/4?	0.083
3 (Bronze)	0.282	0.051
5 (Nickel)	0.310	0.023
10	0.327	0.007
20	0.331	0.002

Monotone from below.

Limit $= 1/3 = \bar{\Lambda}$ of URL cloaked lattice (Klatt et al. 2020).



As $n \to \infty$, tile ratio $L/S \to \infty$, so the chain looks more and more like a lattice with random-looking disorder.

This would match the URL $a = 1$ limit?

Summary

Since our last meeting:

- $\bar{\Lambda}$ from $\sigma^2(R)$ converges fast ($N \sim 10^3$); spreadability needs much larger t for BT
- 3×3 search: 504 chains at α=2 (complex eigenvalue pair forces this); none with $\alpha \in (2, 3)$
- Defined Λ_{II}, Λ_{III} for Classes II/III: Λ_{II}=0.082, Λ_{III}=0.14
- No matrix with $|\det M|$=1 can produce $\alpha \in (2, 3)$ — but $|\det M|$=2 can: first example at α=2.071
- Silver $\bar{\Lambda} = 1/4$, metallic-mean $\bar{\Lambda} \to 1/3$ as $n \to \infty$?

Potential next steps:

- Get more accurate $\bar{\Lambda}$ for the $\alpha = 2.071$ chain (and others) with larger N
- Look for chains with α closer to 3?
- Write up results sections for the JP

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References

- Torquato & Stillinger (2003), Phys. Rev. E **68**, 041113
- Oğuz et al. (2017), Phys. Rev. B **95**, 054119; Oğuz et al. (2019), Acta Cryst. A **75**, 3–13
- Bombieri & Taylor (1986), J. Physique Colloques **47**, C3-19
- Zachary & Torquato (2009), J. Stat. Mech. P12015
- Klatt, Kim & Torquato (2020), Phys. Rev. E **101**, 032118
- Torquato, Zhang & Stillinger (2015), Phys. Rev. X **5**, 021020
- Torquato (2021), Phys. Rev. E **104**, 054102; Wang & Torquato (2022), Phys. Rev. Appl. **17**, 034022
- Hitin-Bialus et al. (2024), Phys. Rev. E **109**, 064108

References

- Torquato & Stillinger (2003), Phys. Rev. E **68**, 041113
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- Torquato, Zhang & Stillinger (2015), Phys. Rev. X **5**, 021020
- Torquato (2021), Phys. Rev. E **104**, 054102; Wang & Torquato (2022), Phys. Rev. Appl. **17**, 034022
- Hitin-Bialus et al. (2024), Phys. Rev. E **109**, 064108